

Final Project Reflection

This project helped me bring together several important concepts from the course, including FastAPI development, database persistence, authentication, frontend integration, Docker, and automated testing. It's nice to see how we started with a basic calculator application and through the semester it has become a more complete full stack web application with user registration, login, protected calculation features, database-backed BREAD functionality, and multiple layers of testing.

One of the biggest things I learned throughout the entire semester was how important it is to keep the backend, frontend, database models, schemas, and tests in mind while expanding the application. After adding the new calculation behavior, I also worked on keeping the operation logic cleaner by centralizing the calculation handling instead of leaving similar logic spread across multiple parts of the codebase. This helped me understand how small refactors can improve maintainability without changing the user-facing features. It also showed me the importance of rerunning unit, integration, and E2E tests after cleanup changes to make sure the application is still working correctly.

This also put into perspective the value of the different levels of testing. Tests helped confirm that individual calculation functions worked correctly, integration tests verified that the API and database behavior worked together properly, and E2E tests helped confirm that the user interface behaved correctly from a user's perspective. When going through the project some of the E2E test failures revealed real backend behavior that needed to be updated. It reminded me of how useful automated tests can be for finding issues that are not always obvious from manual testing.

Overall, this project really strengthened my understanding of full-stack application development and software engineering practices. It gave me more experience organizing a FastAPI project, using SQLAlchemy and Pydantic, protecting routes with authentication, writing tests, and especially debugging failures. I feel as if I will definitely use testing more often in my future projects.